

100,000	200,000	300,000	400,000	500,000	600,000	700,000	800,000	900,000
10,000	20,000	30,000	40,000	50,000	60,000	70,000	80,000	90,000
1,000	2,000	3,000	4,000	5,000	6,000	7,000	8,000	9,000
100	200	300	400	500	600	700	800	900
10	20	30	40	50	60	70	80	90
1	2	3	4	5	6	7	8	9

What number is shown in the Gattegno chart?

Decrease the number shown by 30,000

Increase the number shown by 100,500

Challenge a partner to find other increases and decreases of the number.



Are the statements true or false?

Adding ten thousand to a number only ever changes the digits in exactly one column.

The number consisting of 70 thousands and 400 ones is 700,400

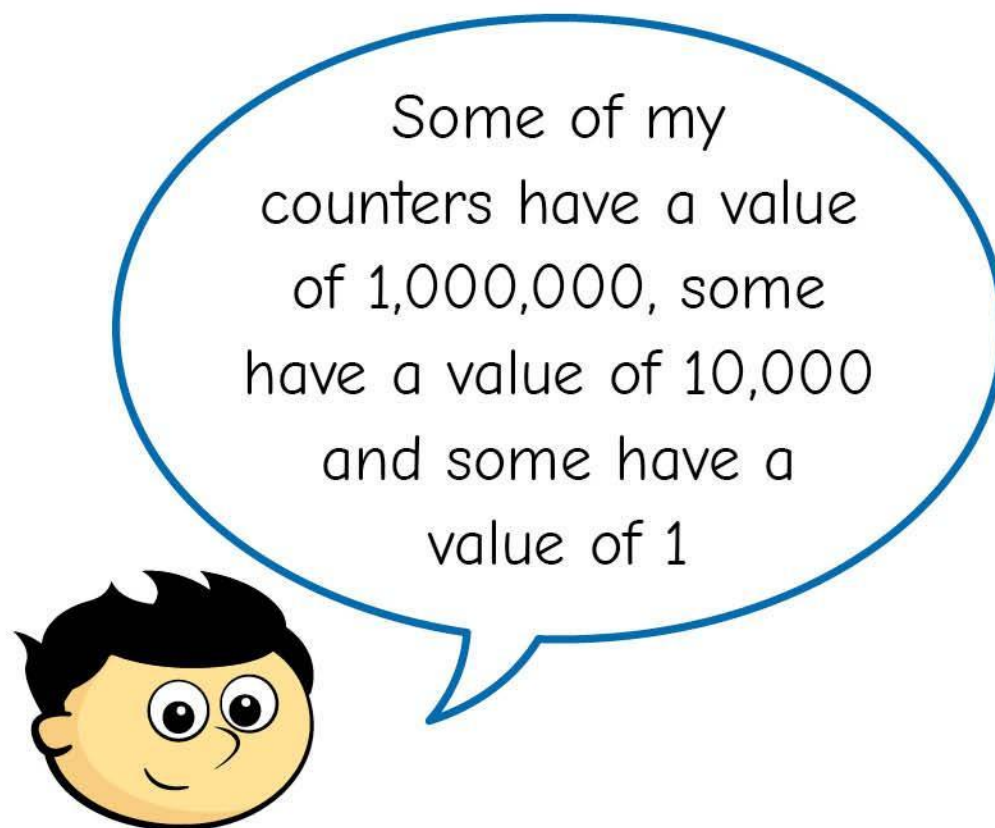
3 ten-thousands is the same as 30 thousands.

400 hundreds is the same as 4 ten-thousands.

A large number added to a large number is always a large number.

A large number subtracted from a large number is always a large number.

Jack has got some place value counters.



Jack picks four counters.

What different numbers greater than 1,000,000 could he make?

Jack wants to make a number greater than 5,000,000

What is the fewest number of counters he needs?

Fill in the missing numbers.

$$824,309 = 800,000 + \underline{\hspace{2cm}} + 4,000 + 300 + 9$$

$$6,413,085 = \underline{\hspace{2cm}} + 80$$

$$58,904 = 50,000 + \underline{\hspace{2cm}} + 4$$

$$947,812 - 400,000 = \underline{\hspace{2cm}}$$

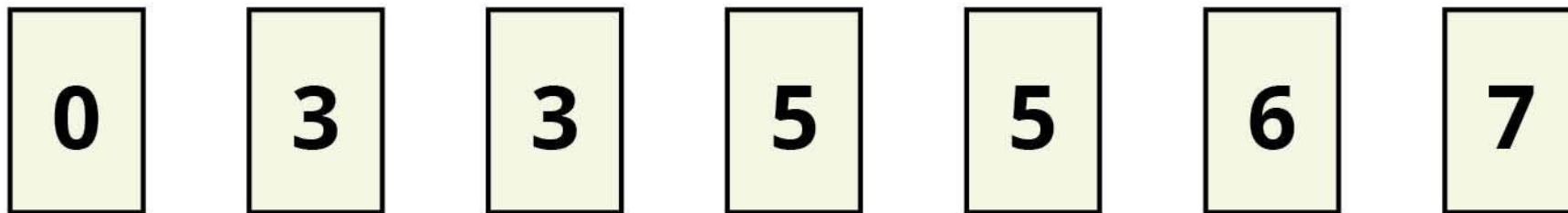
$$947,812 - 4,000 = \underline{\hspace{2cm}}$$

$$947,812 - 400 = \underline{\hspace{2cm}}$$

$$5,198,264 - \underline{\hspace{2cm}} = 5,098,264$$

$$5,198,264 - \underline{\hspace{2cm}} = 5,191,264$$

Use some of the digit cards and the clues to work out the number.



- The ten-thousands and hundreds columns have the same digit.
- The hundred-thousands digit is double the tens digit.
- The number has six digits.
- The number is less than six hundred and fifty-five thousand.

Find as many possible solutions, giving your answers in words and numerals.

Compare answers with a partner.



Here is a number shown on a Gattegno chart.

1,000,000	2,000,000	3,000,000	4,000,000	5,000,000	6,000,000	7,000,000	8,000,000	9,000,000
100,000	200,000	300,000	400,000	500,000	600,000	700,000	800,000	900,000
10,000	20,000	30,000	40,000	50,000	60,000	70,000	80,000	90,000
1,000	2,000	3,000	4,000	5,000	6,000	7,000	8,000	9,000
100	200	300	400	500	600	700	800	900
10	20	30	40	50	60	70	80	90
1	2	3	4	5	6	7	8	9

Write in words the number that is:

- 80 greater than this number
- 80 less than this number
- 80,000 greater than this number
- 80,000 less than this number.

The Gattegno chart shows the answer to a calculation using powers of 10

1,000,000	2,000,000	3,000,000	4,000,000	5,000,000	6,000,000	7,000,000	8,000,000	9,000,000
100,000	200,000	300,000	400,000	500,000	600,000	700,000	800,000	900,000
10,000	20,000	30,000	40,000	50,000	60,000	70,000	80,000	90,000
1,000	2,000	3,000	4,000	5,000	6,000	7,000	8,000	9,000
100	200	300	400	500	600	700	800	900
10	20	30	40	50	60	70	80	90
1	2	3	4	5	6	7	8	9

Find two integer calculations using powers of 10 that give this answer.

Give your answers as calculations, for example:

___ $\times$ (or $\div$) ___ = ___ and sentences such as "___ is 10 times (or one-tenth) the size of ___".

Compare answers with a partner.

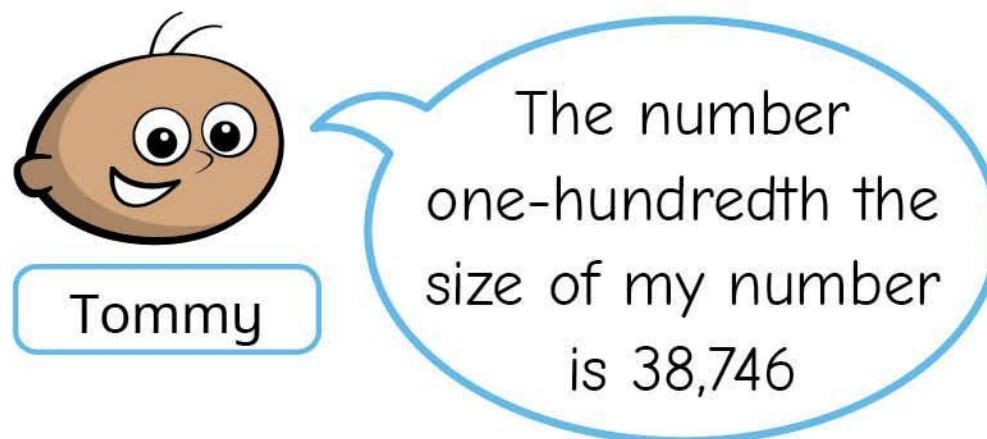


Annie is thinking of a number.



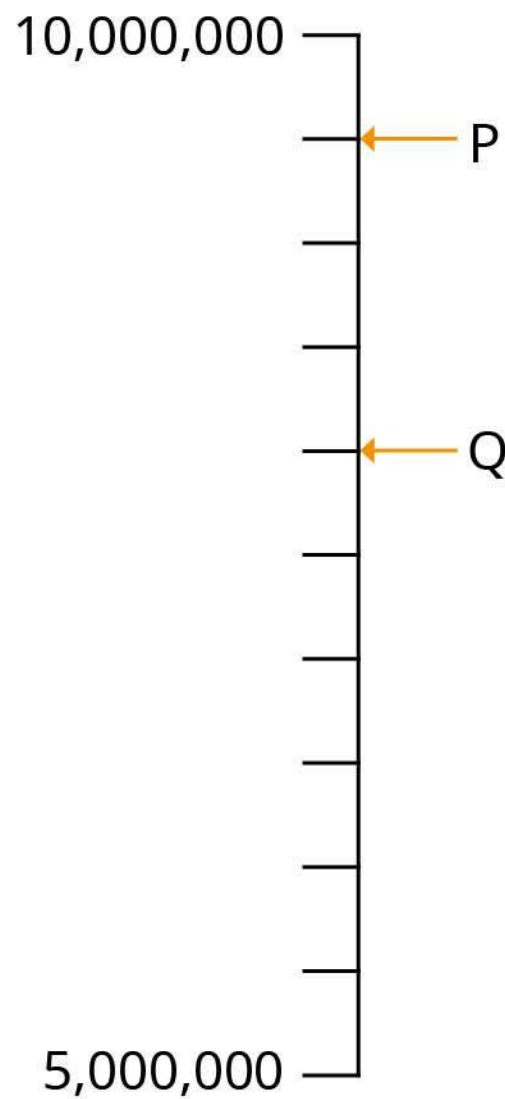
What number is 1,000 times the size of Annie's number?

Tommy is thinking of a number.



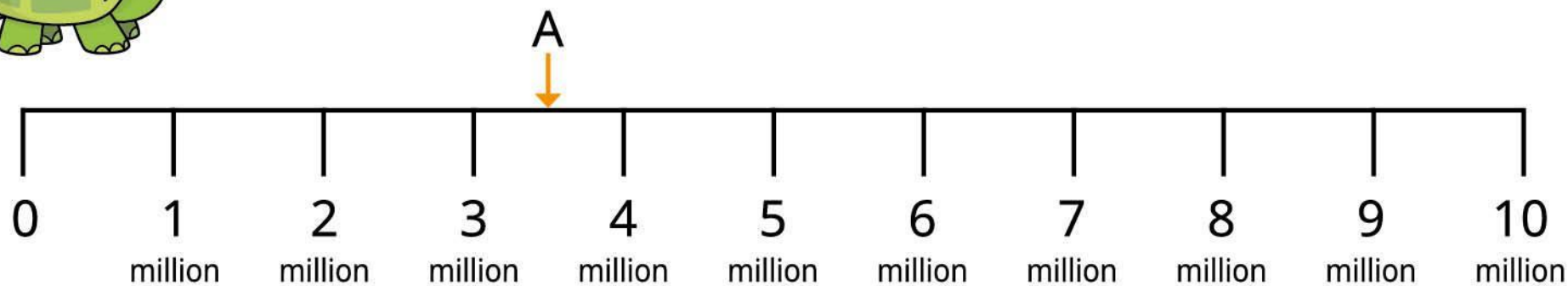
What number is 100 less than Tommy's number?

Find the difference between P and Q.



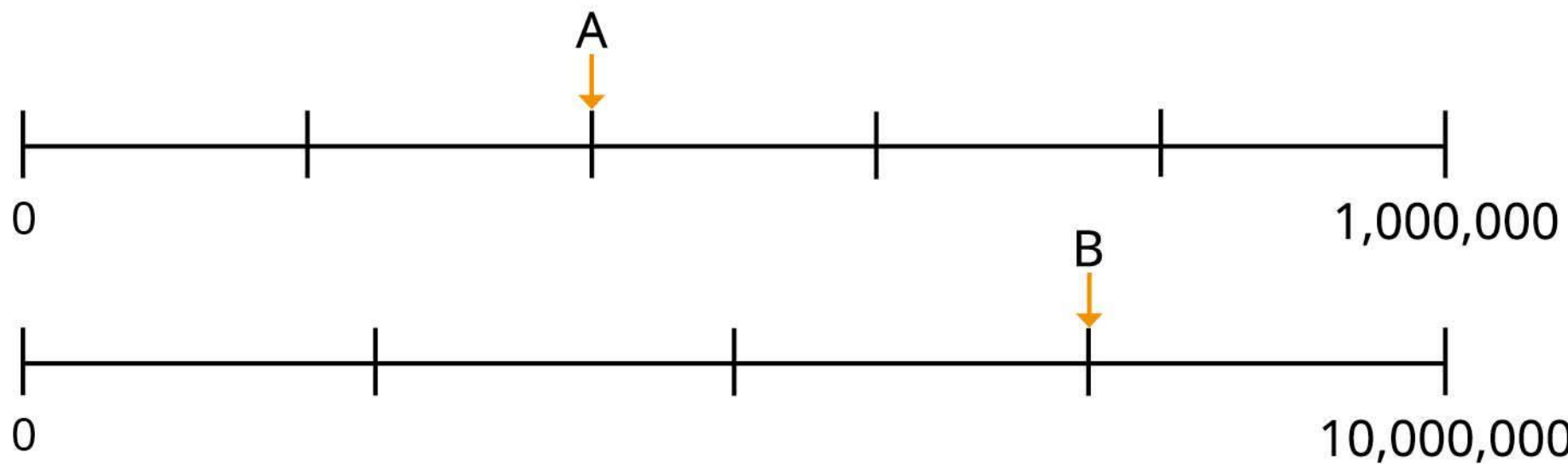
Compare methods with a partner.





Tiny says A is pointing to 3,050,000

Explain the mistake that Tiny has made.



Work out $B - A$.



Eva has put eight 6-digit numbers in ascending order.

- The first number in her list is 345,900
- The last number in her list is 347,000
- All the other numbers in her list have a digit sum of 20
- None of the numbers in her list have any repeated digits.

Find the other six numbers in Eva's list and write them in ascending order.

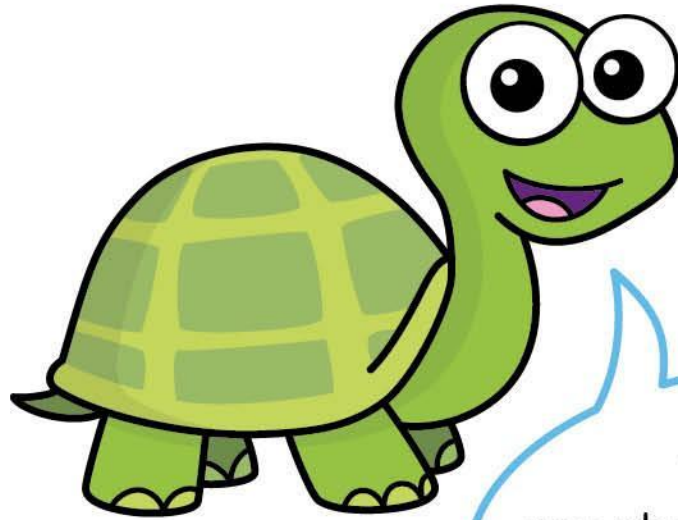
$$\text{_____} + 80,000 < \text{half a million}$$

Complete the sentences.

The missing number could be _____

The missing number cannot be _____

The missing number must be _____

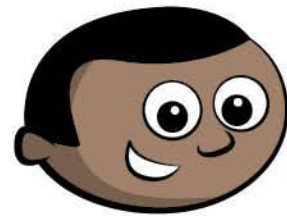


56,700 is
greater than 201,000
because 5 is greater
than 2

Explain the mistake that Tiny has made.

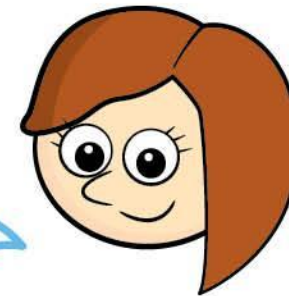


Mo and Rosie are each thinking of a number.



Mo

My number
is 1,350,000 when
rounded to the nearest
ten-thousand.



Rosie

My number
is 1,000,000 when
rounded to the
nearest million.

Both numbers are whole numbers.

What is the greatest possible difference between the two numbers?

Four children each have one of these cards.

15,987

15,813

15,101

16,101

Each child gives a clue about the number on their card.

Filip says, "My number rounds to 16,000 to the nearest thousand."

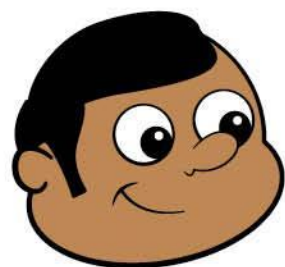
Esther says, "My number has 1 hundred."

Jack says, "My number is 15,990 when rounded to the nearest ten."

Dora says, "My number is 15,000 when rounded to the nearest thousand."

Match the cards to the children.

A company has plans to construct a building with floors above and below ground.



If we build from
floor -10 to floor 10 ,
we will have 20 floors
in total.

Do you agree? Explain your answer.

Find different ways of completing the calculation.

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = -2$$

Is each statement always true, sometimes true or never true?

When you count forwards in tens from a positive 1-digit number, the final digits of all the numbers are the same.

When you count backwards in tens from a positive 1-digit number, the final digits of all the numbers are the same.

Give examples to support your answers.

What patterns can you see?

